



Islamic University of Technology
Department of Computer Science and Engineering

Lab 6: Shortest Path

CSE 4404: Algorithms Lab
Summer 2024-25

Contents

Task A. Arif and Shortest Paths	2
Task Statement	2
Input Format	2
Output Format	2
Constraints	2
Examples	2
Notes	3
Task B. Corporate Network	4
Task Statement	4
Input Format	4
Output Format	4
Examples	4
Notes	4
Task C. Urgent Cables	5
Task Statement	5
Input Format	5
Output Format	5
Examples	5
Notes	5

Task A. Arif and Shortest Paths

Time Limit: 2 seconds | Memory Limit: 256 MB

Arif has a directed weighted graph consisting of n vertices numbered from 1 to n . There is a directed edge between every pair of distinct vertices.

Arif is going to perform a process consisting of n steps. In the i -th step, he selects a vertex x_i and deletes it from the graph along with all its incident edges.

Before deleting each vertex, Arif wants to find the sum of the lengths of the shortest paths between all pairs of remaining vertices. The shortest path can only pass through vertices that have not been deleted yet.

Help Arif find the required sum before each step.

Input Format

The first line contains an integer n — the number of vertices.

The next n lines contain the weight matrix of the graph. The j -th number in the i -th line is the weight of the directed edge from vertex i to vertex j .

The last line contains n distinct integers $x_1, x_2, \dots, x_n$ — the vertices Arif deletes in the order they are removed.

Output Format

Print n space-separated integers — the i -th number should equal the sum of the lengths of the shortest paths between all pairs of remaining vertices before the i -th step.

Constraints

- $1 \leq n \leq 500$
- $1 \leq \text{weight of each edge} \leq 10^5$
- The weight of an edge from any vertex to itself is 0.

Examples

Sample Input	Sample Output
4 0 3 1 1 6 0 400 1 2 4 0 1 1 1 1 0 4 1 2 3	17 23 404 0
2 0 5 4 0 1 2	9 0

Notes

In the first sample, the delete order is 4, 1, 2, 3.

Before deleting vertex 4: all four vertices are present. After running all-pairs shortest paths the sum over the 12 ordered pairs is 17.

Before deleting vertex 1: vertices {1, 2, 3} remain. Vertex 4 can no longer serve as an intermediate. The sum over the 6 ordered pairs is 23.

Before deleting vertex 2: only vertices {2, 3} remain. $d[2][3] = 400$ and $d[3][2] = 4$, giving sum = 404.

Before deleting vertex 3: only one vertex remains, so there are no pairs and the sum is 0.

Task B. Corporate Network

Time Limit: 2 seconds | Memory Limit: 64 MB

Nick's company employs n people, each working at a separate office branch across the city. Nick wants to set up a phone-line network so that any two employees can communicate with each other, directly or through a chain of phone lines.

There are m possible phone lines that can be installed, each connecting two branches at a given monthly cost. Help Nick find the minimum total monthly cost of such a network, or determine that it is impossible to connect all employees.

Input Format

The first line contains an integer n ($1 \leq n \leq 1000$) — the number of employees.

The second line contains an integer m ($0 \leq m \leq 10000$) — the number of possible phone lines.

Each of the following m lines contains three integers a_i , b_i , and c_i ($1 \leq a_i, b_i \leq n$, $0 \leq c_i \leq 10^6$) — indicating that a phone line can be installed between the offices of employees a_i and b_i at monthly cost c_i .

Multiple entries for the same pair of offices are allowed (each at a potentially different cost). All phone lines are bidirectional.

Output Format

Output a single integer — the minimum total monthly cost to connect all employees, or -1 if it is impossible.

Examples

Sample Input	Sample Output
4 4 1 2 5 2 4 1 3 4 1 1 3 5	7
3 2 3 1 2 3 1 3	-1

Notes

In the first sample, the cheapest network installs phone lines between employees $2 \leftrightarrow 4$ (cost 1), $3 \leftrightarrow 4$ (cost 1), and $1 \leftrightarrow 2$ (cost 5), totalling 7.

In the second sample, employee 2 has no available phone-line connections, so it is impossible to reach everyone.

Task C. Urgent Cables

Time Limit: 2 seconds | Memory Limit: 512 MB

Following the cable disaster of July 4202, each damaged cable has been assigned an *urgency rating* by the engineering team — a number reflecting how critical its physical condition is. The repair crew must work through cables strictly in order of urgency (lowest rating first). At each step, they repair the next cable in the list *only if* it connects two currently isolated groups of data centers; otherwise they skip it and move on.

Unfortunately, urgency ratings have nothing to do with repair costs — sometimes the most urgent cable is also the most expensive to fix. Calculate the total repair cost under this urgency-driven schedule, or determine that it is impossible to restore full connectivity.

Input Format

The first line contains two integers n and m ($1 \leq n \leq 10^5$, $1 \leq m \leq 2 \cdot 10^5$) — the number of data centers and cables.

Each of the next m lines contains four integers a , b , c , and p ($1 \leq a, b \leq n$, $1 \leq c \leq 10^9$, $1 \leq p \leq 10^9$) — a cable between data centers a and b with repair cost c and urgency rating p .

All cables are bidirectional, each connects two different data centers, and all urgency ratings are distinct.

Output Format

Print one integer — the total repair cost of the network built by the urgency-driven schedule.

If full connectivity cannot be restored even after processing every cable, print **IMPOSSIBLE**.

Examples

Sample Input	Sample Output
4 5 1 2 10 1 3 4 2 2 2 4 6 3 1 3 8 4 2 3 3 5	18
4 3 1 2 4 3 2 3 7 1 1 3 2 2	IMPOSSIBLE

Notes

In the first sample, cables sorted by urgency are processed as follows: cable 1↔2 (urgency 1, cost 10) — repaired; cable 3↔4 (urgency 2, cost 2) — repaired; cable 2↔4 (urgency 3, cost 6) — repaired, all four data centers are now connected. The remaining cables are never reached. Total cost = 10 + 2 + 6 = 18.

Note that if cables were instead chosen by lowest cost, the optimal spanning tree would cost only 13 — but the urgency schedule does not allow that.

In the second sample, sorted by urgency the order is $2 \leftrightarrow 3$ (urgency 1), $1 \leftrightarrow 3$ (urgency 2), $1 \leftrightarrow 2$ (urgency 3). The first two are repaired; the third is skipped as it would form a cycle. Data center 4 has no available cable, so full connectivity is impossible.

Marks Distribution

Task	Marks
Task A	40%
Task B	30%
Task C	30%

Practice Problems

Problem	Links
Road Reparation	CSES: https://cses.fi/problemset/task/1675
Building Roads	CSES: https://cses.fi/problemset/task/1666
Edges in MST	Codeforces: https://codeforces.com/problemset/problem/160/D
Minimum Spanning Tree	Codeforces: https://codeforces.com/problemset/problem/17/B